Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/02/2022
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Microbial Social Behavior and Heritable Genetic or Epigenetic Changes Affected by the Spaceflight Environment: Understanding the Evolution of Microbial Interactions During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	10/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:	September 2021 report: A lab technician (Sara Koroli, B.S.) worked from October 12, 2020 (Monday) to Friday, May 28, 2021 (Friday). September 2022 report: A student worker Gautam Sai Nakka joined the team but resigned in March 2022 to focus on his course work. Mina Hong (a student worker) has joined the team for this project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1597		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbes naturally exist as mixed-species communities where they actively communicate and interact with neighboring cells. They adapt and evolve over time within the mixed-species populations. While many studies have shown that the spaceflight environment (and ground-based spaceflight analogs) impacts microbial biological processes, most of the previous experiments were based on single-species cultures. It is still unclear how the spaceflight environment influences microbial social behaviors and how their interaction in the mixed-species community contributes to microbial evolutionary processes in the spaceflight environment. Understanding the effects of the spaceflight environment on the changes in microbial social behavior and the associating evolutionary mechanisms will be necessary for spaceflight design, operations, and astronaut health. It is crucial in regard to preparation for long-duration life support systems during future exploration missions far from Earth. Microbes within the ISS are continuously exposed and adapted to the unique spaceflight environment, which includes microgravity, low fluid shear, and enhanced radiation. A variety of microbial isolates have been recovered from the ISS, including the potable water system. The ISS Water Recovery System recycles urine, condensation, and wastewater. Astronauts rely on the recycled potable water for survival. Most of the microbes recovered from ISS portable water are known biofilm formers, which can increase risks for crew infections or damage to life support system function and integrity. Therefore, it is critical to understand how microbes in the ISS water system adapt and evolve over time. The purpose of the study is to understand the microbial evolution associated with interspecies interaction and to examine the influence of the spaceflight environment on microbial interactions and evolutions in the mixed-species population, by cohort profiling. The hypothesis is that changes in microbial interaction between species of ISS water isolates are associated with microbial genomic evolution during spaceflight, and spaceflight is a contributing factor for the evolution of microbial interaction between different species.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is searching for an understanding by which microorganisms interact and evolve in the context of the complex microbial ecosystem. While immediately applicable to the unique spaceflight environment, it will also be applicable to closed environments in which water can be recycled and reused in terrestrial environments. Overall, this contributes to the creation of healthier and sustainable human built and living environments both in space and on Earth. The study uses the International Space Station as a microbial observatory to study the complex microbial ecosystems and dynamics in human built and living environments, including spaceflights. The study aligns with the research goals in the Space Biology Microbiology Element to determine the effect of spaceflight on life, process, and community dynamics. Completion of this project would 1) provide genomic evidence on the potential microevolution of ISS potable water, 2) provide insight into how microorganisms in the heterogeneous population evolve over time, and 3) determine whether the spaceflight environment is a key contributing factor that influences changes in microbial interaction. The knowledge gained from this study may a) fill knowledge gaps in microbial interspecies dynamics and evolution within the mixed-species population; b) provide novel insight into how the microbial ecological success within the mixed-species population contributes to microbial evolution in human Built Environment, whether in spaceflight or on Earth; and c) provide useful knowledge to support long-duration life support system, particularly, a water recovery system in spaceflight.
Task Progress:	This is the 3rd year project report. Our progress: 1. We completed 4 oxygen tests. 2. We successfully obtained groundwater (Earth water) isolates, completed growth profiling, and flight readiness tests prior to the Science Verification Test (SVT). 3. The whole genome sequences (WGS) of the Earth water isolate Sphingomonas spp was assembled. We expect to make the sequence analysis public soon. 4. We also received 16 new strains isolated from the International Space Station (ISS), from the NASA Johnson Space Center (JSC). All ISS Sphingomonas that we received from the JSC were sequenced at the Arizona State University (ASU) sequencing center. The WGS of each strain were assembled and the assembled sequences are being analyzed. We will report the full genome sequence data and the sequence analysis as early as it can be completed. 5. The interactive relationships between select species isolated from the ISS in different years are being tested with various combinations. 6. Science Verification Test: SVT procedure development, Colony Forming Units (CFU) training, test plan development and document release, three preparation attempts, contamination check, hardware assembly and experiment loading, problem assessment, and troubleshooting were completed prior to inoculation. a. SVT was initiated in August 2022 and was terminated early due to hardware failures across all 3 cassettes, 21 days after inoculation. Prior to the complete hardware failure, two Mix A fluid loops leaked and were removed from the science evaluation. An investigation is underway by the hardware developer with the intention of redesigning particular hardware components to increase space in the cassettes and minimize pinch points. The schedule impact is unknown at this time.
Bibliography Type:	Description: (Last Updated: 09/02/2025)
Articles in Other Journals or Periodicals	Velez Justiniano Y-A, Goeres D, Sandvik E, Kjellerup BV, Sysoeva T, Harris J, Foreman C, McLean RJC, Yang J, Li W, O'Rourke A, Cassilly C, Lott K, HerrNeckar L. "Mitigation and use of biofilms in space for the benefit of human space exploration." National Academies of Sciences, Engineering, and Medicine. Topical White Paper. To appear in special decadal survey on biological and physical sciences research in space 2023-2032. In review. http://surveygizmoresponseuploads.s3.amazonaws.com/fileuploads/623127/6378869/106-32e9a4f2009e67a2765c69a8bf3002c8_VelezYo-Ann_rev2.pdf , Jun-2023
Articles in Other Journals or Periodicals	Santomartino R, Averesch NJH, Bhuiyan M, Cockell CS, Colangelo J, Gumulya Y, Jha D, Lehner B, Lopez I, McMahon S, Mohanty A, Santa Maria SR , Urbaniak C, Volger R, Yang J. "Biologically facilitated processes towards sustainable space exploration." National Academies of Sciences, Engineering, and Medicine. Topical White Paper. To appear in special decadal survey on biological and physical sciences research in space 2023-2032. In review. http://surveygizmoresponseuploads.s3.amazonaws.com/fileuploads/623127/6378869/143-04a1ffaf4358b71a08fc883758723d79_SantomartinoRosa.pdf , Jun-2023